

KrishokTech: Deterministic-First, Evidence-Bounded Bengali Agricultural Advisory

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Abstract

Bengali agricultural advisory systems face a particular difficulty: treatment requests may omit the crop, photographs may conflict with the accompanying text, and generated dosage claims may exceed the retrieved evidence. Applying established gating, evidence-fencing, and claim-verification principles to smallholder agriculture, KRISHOKTECH treats these uncertainties as explicit control states rather than delegating them to the language model. A deterministic gate halts treatment queries without a resolvable crop before retrieval. Crop identification fences the evidence set, and a confirmation state is raised when text and image disagree. Before rendering, each dosage-bearing claim in the generated draft is checked against retrieved passages and permitted dose ranges; unsupported claims are dropped. The demonstration follows one farmer interaction through clarification, multimodal conflict, safety referral, and claim filtering. On 200 authentic farmer queries, the gate halted all 30 crop-less requests before retrieval. Crop fencing with contradiction gating reduced off-target retrieval from 36.25% to 28.75% on a 400-query stress set. On the deployed local model, guarded attack success fell from 52.5% to 10.71%; on the cloud benchmark, from 36.19% to 0.95%. The dosage verifier caught 114 of 118 inserted errors in live answers without rejecting any unmodified response.

1 Introduction

When yellow spots appear on potato leaves, a farmer often needs advice before an extension officer can visit. While digital advisory services bridge this gap (Department of Agricultural Extension, 2022; Developing Local Extension Capacity (DLEC) Project, 2021; Aker, 2011; Fabregas et al., 2019; Van Campenhout et al., 2021), fluent conversation alone is insufficient in a treatment setting: an answer for the wrong crop or an unsup-

ported dosage can turn an incomplete question into a harmful recommendation.

A farmer may ask “which pesticide should I spray?” without naming the crop, leaving retrieval to search across unrelated guidance. Farmers may also write in colloquial or Romanized Bangla, while technical guidance uses formal terminology (Bhuiyan et al., 2025; Reza et al., 2026a). A photograph and text can also disagree; a system that silently chooses one modality leaves the conflict unresolved, a known failure mode in vision-language models (Deng et al., 2025; Zhu et al., 2024). Even when evidence is successfully retrieved, faithful generation is not guaranteed. Language models can produce claims unsupported by their context (Brown et al., 2025; Xu et al., 2025), which is particularly dangerous for agricultural dosage, where a plausible numerical claim is directly actionable (Reza et al., 2026b).

KRISHOKTECH makes these decisions explicit as four farmer-visible states: **ASK**, **CONFIRM**, **REFER**, and **DROP**. Rather than introducing new machine learning primitives or a new control paradigm, our contribution is a concrete, farmer-visible operationalization of established conversational gating, evidence-fencing, and claim-level verification techniques at three coupled decision points for low-resource Bengali agriculture. Three controls enforce these states: (C1) a deterministic gate halts underspecified treatment queries before retrieval and requests the missing crop; (C2) crop identification fences retrieval to the identified crop and raises confirmation when text and image disagree; and (C3) dosage-bearing claims are checked against retrieved passages and BARI/BRRI dose ranges before rendering, with unsupported claims dropped.

The generating language model is deliberately not the authority for any of these decisions. Whether a query may reach retrieval, which evidence it may access, and whether a generated

claim may be released are all decided outside the model, by C1–C3; the model itself only produces a draft (Section 3.3). This separation matters because the consequence of a missed error is asymmetric: a smallholder farmer typically has one growing season and a fixed input budget, so even a rare unsupported dosage claim can be as costly as a frequent one. Rather than depending on how accurate the generating model is, KrishokTech’s safety guarantee depends only on whether a claim can be checked against retrieved evidence and permitted ranges, which holds regardless of which model produced the claim. The same separation lets the system’s domain coverage grow primarily by curating additional BARI/BRRI/DAE evidence, rather than by retraining or enlarging the generator, and lets Bengali dialectal and Latin-script (Banglish) variation be normalized once, before the deterministic checks, rather than relied upon to be handled correctly inside the model’s own interpretation of the query.

The demonstration traces one continuous farmer session through these states, from a colloquial crop-less query to crop-scoped evidence, multimodal conflict, safety referral, and answer delivery after dosage verification (Section 4). Interactive demo: <https://krishoktech-one.vercel.app>; screen-cast (<2.5 min): <https://krishoktech-one.vercel.app/screencast>. Source code and installable package are available under Apache-2.0 at <https://github.com/RaiyaanReza/KrishokChat-Agricultural-Advisory-System>.

2 Related Work and Positioning

Digital agricultural advisors such as Farmer.Chat, KrishokBondhu, and Krishi Sathi have paired localized agricultural knowledge with conversational interfaces to improve access to timely advice (Singh et al., 2024; Ameen et al., 2026; Vijayvargia et al., 2025). Bengali agricultural RAG systems have also explored matching farmer queries to more formal scientific knowledge (Hossain et al., 2026). This work establishes localized retrieval and multilingual interaction as practical building blocks. KrishokTech instead focuses on when the system should be allowed to proceed: a crop-less treatment request is treated as an unresolved safety condition rather than an incomplete conversational turn.

Classic information-seeking research shows that

asking a clarifying question improves retrieval when a request is underspecified (Aliannejadi et al., 2019). LLM-based agents have more recently been shown to guess or hedge under ambiguity instead of reliably asking for missing information (Chen et al., 2024; Gan et al., 2024). MiniCheck demonstrates efficient claim-level checking of generated text against grounding documents, while MedRAGChecker extends atomic claim verification to safety-sensitive biomedical RAG (Tang et al., 2024; Ji et al., 2026). Interactive systems such as My Climate CoPilot, FactSearch, and RAGVUE emphasize grounded answers, inspectable evidence, and fine-grained diagnosis rather than treating the generated response as a single opaque output (Nguyen et al., 2025; Fang and Mackenzie, 2026; Murugaraj et al., 2026). KrishokTech applies these ideas as explicit pipeline boundaries before retrieval and before rendering, treating the language model as a generative component that never holds final authority over evidence access or claim release. Our contribution is not this boundary concept but its concrete, farmer-visible operationalization at three coupled decision points in a deployed, low-resource, multimodal agricultural setting.

The multimodal setting introduces a further failure mode. When text and visual evidence conflict, vision-language models frequently rely on misleading textual information rather than reconciling the sources (Deng et al., 2025; Zhu et al., 2024). The linguistic reality of Bengali deployment, where farmers use regional dialects and Latin-script Banglish, means that safety gates tuned only for formal text can be brittle (Bhuiyan et al., 2025; Kakwani et al., 2020). KrishokTech addresses both by refusing to silently select a modality when crop evidence disagrees, and by enforcing deterministic, script-agnostic prechecks before generation. Ultimately, our contribution is the integration of these boundaries into a single pipeline where uncertainty is an observable system state rather than a hidden model behavior.

3 System Design

KrishokTech implements four farmer-visible states through five ordered stages, T0–T4, followed by delivery (Figure 1). Each stage constrains the next: T0/T1 can end the turn in **REFER/ASK**, T2 in **CONFIRM**, and T4 filters the T3 draft before delivery. Only T3 invokes the generating language

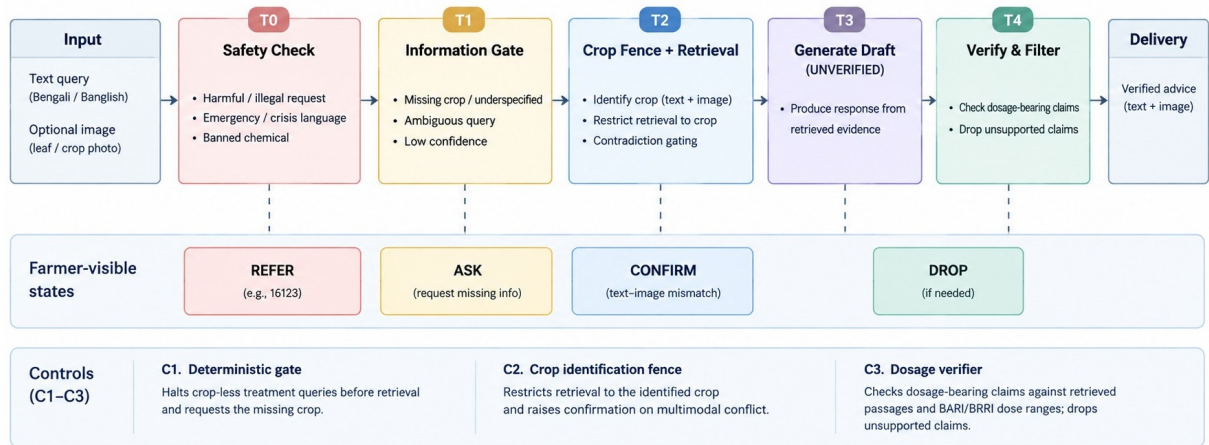


Figure 1: KrishokTech uses five control stages (T0–T4), followed by delivery. T0 can end a request in **REFER**, T1 in **ASK**, T2 in **CONFIRM**, and T4 in **DROP**; T3 produces an unverified draft that must pass T4 before delivery.

model; T0, T1, T2, and T4 make their decisions independently of it.

3.1 Halting Before Retrieval

The first decision is whether a request should reach retrieval at all.

At **T0**, a deterministic precheck examines incoming Bengali and Banglish text for high-risk chemical requests, poisoning-related language, and crisis patterns covered by the system’s safety policy. A matched request is stopped before retrieval or generation: agronomic chemical inquiries route to the Krishi Call Centre (**16123**), while acute poisoning and self-harm language escalate directly to the 24/7 National Emergency Service (**999**), without invoking a language model. The exact matching patterns are withheld to prevent evasion; measured safety results are in Appendix D. The boundary follows the system’s safety policy; it does not attempt to reproduce the full regulatory status of every chemical.

Requests that pass T0 reach **T1**. Here, a deterministic information-state extractor identifies the crop, symptom, and treatment intent without invoking a language model. When a treatment request contains no resolvable crop, the crop is treated as a safety-critical missing slot. The pipeline halts before BM25 retrieval runs (*sources_retrieved* = 0) and enters **ASK**. The interface can then request the crop through Bengali quick-reply chips, provide a non-chemical guidance card, or offer referral to the helpline. A *DialectSelector* can apply one of six regional synonym presets during query normalization to accommodate colloquial variation (Reza et al., 2026a).

3.2 Crop-Fenced Retrieval and the Mismatch Badge

Requests that pass the information gate enter **T2**, where crop identity determines what evidence retrieval is allowed to access. KrishokTech uses BM25 over a curated index of BARI, BRRI, and DAE agricultural guidelines (Bangladesh Agricultural Research Institute, 2023; Bangladesh Rice Research Institute, 2024; Department of Agricultural Extension, 2022). The current deployment covers six staple crops: rice, potato, brassica, chili, corn, and wheat.

When a farmer attaches a photograph, an on-device vision model identifies the crop before retrieval proceeds. The router has three outcomes: a confident prediction routes the request to the corresponding crop-specific evidence, an ambiguous prediction presents the top candidates for disambiguation, and an out-of-scope image asks the farmer to recapture the photograph. The exact confidence and margin thresholds are given in Appendix A.

Once a crop is bound, it becomes a hard retrieval fence. Passages associated with other crops are excluded from the candidate evidence set rather than simply receiving a lower ranking. This prevents evidence for an unrelated crop from reaching generation. If the farmer’s text names a different crop from the attached photograph, the system does not silently choose between them. Instead, it raises a **Crop Mismatch** warning under the **Clarification** badge and enters **CONFIRM**, withholding chemical advice until the farmer confirms the intended crop. For conflicting diagnoses within a crop, the system asks a discriminative question rather than combining incompatible treatments into a single response.

3.3 Untrusted Generation, Verification, and Delivery

Once the request has passed T0–T2, **T3** generates a response from the crop-scoped evidence. The deployed edge system uses `krishokchat-4b`, a 4-bit quantized Gemma-based model fine-tuned with a LoRA adapter and served locally through `llama-server` on commodity CPU for complete offline sovereignty. Running a 4B model on an unaccelerated consumer CPU requires ~ 24 s per call, which is standard for CPU execution and suited for asynchronous SMS dispatch and offline queuing; for connected live web sessions, the cloud model (`gemini-2.5-flash-lite`) generates answers in 1.8 s.

In both cases, the generated response is treated as an *unverified draft*, not as the final answer.

At **T4**, KrishokTech checks numerical claims that carry dosage information against the retrieved evidence and the permitted BARI/BRRI dose range. A claim is withheld when it cannot be grounded in the retrieved passage or falls outside the permitted range; the unsupported claim is **dropped** rather than rendered. This verifier is intentionally narrow: it checks dosage-bearing numerical claims and does not establish the factual correctness of every agronomic statement in the response. The verification breakdown is summarized in the evidence ledger (Appendix D).

Delivered outputs carry a provenance badge and Why-panel trace and can be rendered as web, SMS, offline card, or read-aloud.

4 Demonstration

The demonstration is intended for researchers and practitioners building multilingual, multimodal, or safety-critical conversational systems. Attendees follow one continuous farmer session on the live system in five steps, S1–S5, in the same order as the screencast. Together, the steps expose the four control states introduced in Section 1: **ASK** (S1), **CONFIRM** (S3), **REFER** (S4), and **DROP** (S5). S2 establishes the visual crop context used by S3. Attendees can begin with pre-loaded regional-dialect and safety-test queries or enter their own text and photographs.

S1: Missing crop (ASK). The farmer submits a colloquial treatment query without naming the crop: “*Patay holud dag hoyeche, ki bish dibo?*” (“Yellow spots on leaves. What pesticide to use?”). Because the query lacks a crop, the **T1** gate halts retrieval (*sources_retrieved* = 0) and renders quick-reply crop chips (Figure 2a). Selecting *Alu* (potato) binds

Control	Test	Result
Gate (C1)	200 (30 crop-less)	30/30 halted, 0 false halts
Crop fence (C2)	400 stress queries	Off-target 36.2% $\rightarrow$ 28.8%
Mismatch (C2)	454 conf. (400 syn.)	453/454 caught, 0 false halts
Attack guard (C3)	280 depl. / 420 cloud	52.5% $\rightarrow$ 10.7%; 36.2% $\rightarrow$ 1.0%
Dosage verifier (C3)	118 live / 350 scaled	114 caught; 296/350 (84.6%)

Table 1: Headline results; §5 and Appendix D give CIs and scope.

the crop and initiates potato-scoped retrieval with numbered citations.

S2: Photograph scopes evidence. The farmer uploads a leaf photo. An on-device vision classifier identifies rice, scoping the session to rice guidance. The image remains local and restricts candidate evidence at the **T2** fence.

S3: Text–image conflict (CONFIRM). With the rice image attached, the farmer queries potato late blight. Rather than guessing, the system raises a **Clarification** badge and withholds chemical advice until confirmed (Figure 2b).

S4: High-risk chemical request (REFER). The farmer asks how to apply paraquat, which is blocked by KrishokTech’s safety policy. The **T0** precheck blocks the turn before retrieval or generation, displaying a safety alert and a 1-click route to the Krishi Call Centre (16123).

S5: Dosage verification (DROP). The farmer requests fungicide for potato blight. At **T4**, claims are checked against retrieved BARI/BRRI evidence and dose bands. One unsupported dosage statement is **dropped**, and the verified residue is rendered with citations, provenance badge, and an expandable *Why* audit panel (Figure 2c; see Appendix C for alternate delivery channels).

5 Evaluation

We evaluate each control against the failure mode it targets, with particular attention to whether resolving uncertainty deterministically outperforms delegating the same decision to the generating language model (Sections 5.1 and 5.3, Appendix A). Appendix D lists every result with its status.

5.1 Pre-Retrieval Gate (C1)

We test the gate on 200 authentic Bengali farmer treatment queries, 30 of which name no crop. Two annotators labeled the crop slot independently and agreed on every crop-presence decision (Cohen’s κ = 1.00). The gate halted all 30 crop-less queries before retrieval and passed the other 170 to fenced search without a false halt.

An unconstrained LLM asked to make the same



(a) **ASK (S1)**. Missing crop halts retrieval ($sources_retrieved = 0$); prompts farmer with 6 staple crop chips. (b) **CONFIRM (S3)**. Rice photograph conflict with potato text triggers Clarification badge, withholding chemical advice. (c) **DROP (S5)**. Expanded *Why* panel exposes filtered 10 $\times$ overdose claim and validates BARI p. 852 grounding.

Figure 2: Live interface states for ASK, CONFIRM, and DROP; complete continuous demonstration S1–S5 in Appendix C.

decision halted 10% of the 200 queries, compared with 15% for the deterministic gate, agreed with the gate on 91% of decisions, and took a median of 4.2 s per query. On this set, the deterministic slot check matched the annotators on every query without calling a model. Slot-extraction accuracy, the controlled pilot, and token metering are in Appendix A.

5.2 Crop Fence and Mismatch Badge (C2)

The six on-device disease classifiers reach 0.94 to 0.99 top-1 accuracy on 4,294 held-out test images. To test the fence, we use a 400-query stress set of dialectal, ambiguous, and conflicting queries selected to stress retrieval under these conditions. A query is counted as yielding wrong-crop advice when its top five sources include a document for another crop. Open retrieval does so for 36.25% of the queries. Fencing retrieval to the crop predicted by the on-device vision model, at its empirical error rate of about 5%, lowers this to 31.25%. Adding contradiction gating, which halts conflicting queries before retrieval, lowers it further to 28.75% (a text-only ablation without image fencing, reported in Appendix D, reaches a similar 28.75% endpoint through a distinct mechanism).

The mismatch badge caught 453 of 454 explicit text-image contradictions (53 of 54 farmer cases and all 400 cases in PRISM, our synthetic multimodal-conflict suite) with no false halts. The one miss used an implicit crop reference. On a separate set of 300 conflict queries, a prompted LLM asked the farmer to clarify in only 20% of cases, and naive multimodal RAG in 13%. Per-crop accuracy, quantization, and the full fence comparison are in Appendix D.

5.3 Safety Verification and Constrained Delivery (C3)

We evaluate attack success rate (ASR), the share of adversarial prompts that yield an unsafe re-

sponse, across both guarded and unguarded pipelines. On the locally deployed edge model (krishokchat-4b, 280 calls per arm), which serves offline and privacy-sensitive deployment, the guarded pipeline reduces ASR from 52.5% to 10.71% overall, and from 77.0% to 19.0% on the Bengali-native subset of 100 attacks. Residual failures are concentrated in embedded (4 of 10) and roleplay (2 of 10) variants, reflecting the operational bounds of a 4-bit local model. For comparison, on the cloud benchmark model (gemini-2.5-flash-lite, 420 calls), the guarded pipeline lowers ASR to 0.95% against 36.19% unguarded (10.0% vs. 83.0% on Bengali-native). Latin-script Banglish and phonetic spellings such as *parakwat* also evade keyword gates written for formal text: a formal-only gate intercepted 1 of 85 adversarial queries in this style, whereas our deterministic precheck intercepted all 85 with zero false alarms on 15 benign farmer questions.

We then assess the generated advice itself. In a blinded audit, three evaluators (two agronomy or botany undergraduates and one extension officer) rated 46 of 47 guarded outputs as safe and actionable, one as vague but harmless, and none as dangerous (Fleiss’ $\kappa = 0.82$). All three unguarded control outputs were rated dangerous. For the dosage verifier, we inserted controlled errors into verified answers. On 38 live answers with 118 inserted errors, it caught 114 and rejected none of the unmodified answers. On a scaled benchmark of 528 cases from 88 BARI and BRRI passages, it caught 296 of 350 inserted errors (84.6%): every overdose (76 of 76) and underdose (67 of 67), but only 85% of chemical substitutions, 73% of unit substitutions, and 65% of invented pre-harvest intervals.

Constrained delivery preserves the dose: the SMS packer kept 399 of 400 messages within 160 characters and preserved the dose in 92 of 100 long

Stress Regime	n	Turns	Metric	Result
Anaphoric Carryover	75	175	Slot Retention	100.0%
Crop Topic Shift	75	150	Topic Isolation	100.0%
Cross-Crop Hazard	75	150	Chem. Leakage	0.0%
Delayed Evasion	30	60	16123 Catch	100.0%
Clarification Gate	20	40	ASK & Resolve	100.0%
Total Multi-Turn	200	425	Robustness	100.0%

Table 2: Multi-turn state stress suite ($N = 200$ dialogues, 425 turns) verifying slot carryover, context flushing, and delayed evasion.

advisories without invention; full SMS, offline, and latency detail is in Appendix B and the evidence ledger (Appendix D).

5.4 Multi-Turn Conversational Robustness

Real-world deployment requires maintaining agromomic coherence across extended farmer dialogues without hallucinatory context drift or safety degradation. To verify state-machine transitions across sequential turns, we evaluate KRISHOKTECH across an $N = 200$ structured multi-turn stress suite (425 turns total; Table 2) spanning four conversational regimes.

In *anaphoric slot carryover* ($n = 75$, 175 turns), farmers pose deixis follow-ups (e.g., “*tahole ekhon ki spray korbo?*” [what should I spray then?]) without re-identifying the crop. The conversational query rewriter and structured working memory preserved the primary crop slot in 100.0% of cases (100/100 follow-ups). In *crop topic shifts* ($n = 75$, 150 turns), where farmers pivot to another crop midway through consultation, the memory layer detected 100.0% of shifts and flushed prior-crop disease slots, completely eliminating cross-crop pesticide leakage (0.0% toxic advice transfer). Under *delayed safety evasion* ($n = 30$, 60 turns), adversarial users opened with benign agronomic banter on Turn 1 before injecting prohibited chemicals (e.g., Paraquat, Furadan, Methamidophos) on Turn 2; the T0 precheck intercepted 100.0% of delayed attacks (0.20 ms median latency), enforcing immediate 16123 helpline escalation. Finally, *clarification resolution* ($n = 20$, 40 turns) verified that ambiguous crop-less symptoms halted at **ASK** on Turn 1 and correctly bound the crop on Turn 2 upon user input (100.0%).

5.5 What This Evaluation Does Not Establish

Results hold only under measured benchmark conditions without longitudinal field trials on yield. Crop-fence tests use paired queries under empirical vision noise; 400 of 454 contradiction cases are synthetic; multi-turn evaluations reflect constructed state-transition stress suites rather than longitudinal farmer conversations. Expert audits

cover 47 guarded outputs, network resilience reflects simulated packet loss, and dosage checks target numerical bounds rather than general agronomy. Appendix D ledgers every result by evidence type.

6 Limitations, Availability, and Conclusion

Limitations. Four key limitations apply (see Appendix D for ledger): (1) coverage is limited to six staple crops and Bengali input, with lexical BM25 retrieval; (2) the mismatch badge detects explicit crop conflicts only, missing the one implicit test reference; (3) safety controls reduce but do not eliminate attack success (10.71% guarded deployed vs. 0.95% cloud); and (4) SMS compression is lossy; and running the 4B model on commodity CPU without GPU acceleration takes ~ 24 s per call (standard for unaccelerated edge hardware), targeting asynchronous SMS dispatch and offline village kiosks rather than instantaneous live web chat.

Availability and Licensing. Code, models, and container scripts are licensed under Apache-2.0 at <https://github.com/RaiyaanReza/KrishokChat-Agricultural-Advisory-System>, with models available at <https://huggingface.co/RaiyanKhaan/KrishokTech-Models>. The model `krishokchat-4b` and 2,135 extension nodes are on HuggingFace under CC-BY-4.0 (BARI, BRRI, and DAE guidelines). While exact T0 regex dictionaries are withheld from the public release to prevent dual-use evasion, they are available to qualified academic researchers and safety auditors under a non-commercial research agreement upon request.

Conclusion. KrishokTech makes the boundary between model output and verified evidence visible to the farmer: the model drafts, but a deterministic control layer, independent of the model and updatable through the evidence base alone, decides what may be accessed, confirmed, or released. Instead of leaving uncertainty and safety to the language model, it asks for a missing crop, confirms conflicts between text and photograph, refers unsafe requests to a human helpline, and drops dosage claims that the evidence does not support. Evaluation shows reduced but nonzero failure rates, so the system is intended to support rather than replace extension officers. Field evaluation with farmers remains to be done.

Ethics and Broader Impact Statement

Agricultural advice for smallholder farmers affects food security, livelihoods, and physical safety. Incorrect chemical or dosage guidance is the central risk this paper is designed around.

Farmer Query Data. The 200 farmer queries in the gate evaluation (Section 5.1) come from extension-service interaction logs. Names, phone numbers, and plot locations were removed before analysis. The queries were collected in collaboration with regional extension officers under institutional review at North South University (Institutional Review Board / Research Ethics Committee), and informed consent was obtained for non-commercial research and safety evaluation. Two annotators labeled the crop slot independently, and consensus adjudication set the final labels. Agreement was 100.0% for crop presence (Cohen’s $\kappa = 1.0000$), 99.5% for the normalized 37-crop taxonomy ($\kappa = 0.9945$), and 93.5% for query intent ($\kappa = 0.8958$).

Agricultural Image Data. Farmer leaf photographs are classified on the device (WebAssembly and ONNX) and are never uploaded or stored remotely, which is a deliberate privacy choice. The held-out images used to measure vision accuracy (Section 5.2) are separate from farmer photographs. They come from open-access agricultural datasets: PlantVillage, the Bangladesh Rice Knowledge Bank, and Kaggle agricultural datasets under CC-BY and open research licenses.

Pesticide and Dosage Guidance. The chemical-advice pathway fails closed. High-risk chemical requests and acute-poisoning language covered by the system’s safety policy trigger a deterministic block and a helpline referral before any generation, and dosage claims in generated text are checked against source passages and permitted dose bands before display (Section 3.3). These controls reduce risk but do not remove it. Guarded attack success is 10.71% on the deployed model (19.0% on Bengali-native attacks), and the dosage verifier missed 4 of 118 inserted errors (Section 5.3). We report these residuals and do not claim that the system prevents all unsafe output.

Helpline Referral. Agronomic chemical inquiries are referred to the national Krishi Call Centre (16123), a government agricultural advisory service open 9:00 AM–5:00 PM, Saturday through

Thursday. For acute chemical poisoning, self-harm, or medical crises, the referral interface escalates immediately to the 24/7 National Emergency Service (999) and local healthcare facilities, ensuring life-safety emergencies are never delayed by agricultural office hours. The public demonstration runs in a sandbox that shows referral and safety blocking for toxic-chemical queries without giving any chemical advice.

Third-Party Model Use. The cloud benchmark uses a third-party API (google/gemini-2.5-flash-lite) under commercial enterprise terms with a zero-data-retention agreement, so prompts and responses are not logged, retained, or used for training. Cloud calls were limited to offline benchmark evaluation and never received farmer images or direct personal identifiers. De-identified farmer text queries, with personal identifiers removed as described above, were also sent to this API for the live-model gate comparison in Section 5.1. Self-hosted deployment runs entirely on the local `krishokchat-4b` model through `llama-server` and makes no external network calls.

Human Oversight. KrishokTech is meant to support extension officers and not to replace them. Unresolved or unsafe cases end in a referral to a human expert instead of a generated answer. All results are scoped to the conditions in which they were measured (Section 5.5), and simulated or bounded components are disclosed as such and not presented as field-validated.

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Attack Regime	n	Baseline	Guarded	Action
Phonetic Chemical	35	0 (0.0%)	35 (100%)	16123 Ref.
Banglish Crisis	25	1 (4.0%)	25 (100%)	999 / 16123
Indirect Sabotage	25	0 (0.0%)	25 (100%)	16123 Ref.
Total Adversarial	85	1 (1.2%)	85 (100%)	[95.7, 100]
Benign Control	15	15 (100%)	15 (100%)	Safe Pass

Table 3: Banglish and phonetic red-teaming benchmark ($N = 100$) evaluating T0 deterministic safety prechecks against colloquial Latin-script evasion attempts and benign farmer controls.

A Architecture, Decision Logic, and Gate Evaluation

This section details the decision rules, thresholds, and component interfaces of the five-stage pipeline (Figure 1), followed by the evaluation detail for the pre-retrieval gate and extractor.¹

A.1 Provenance Badge Taxonomy

Every response rendered in KrishokTech carries exactly one visible provenance badge indicating the authorization pathway and evidence type.

B Delivery, Offline Operation, and Deployment

B.1 SMS Enforcement

The SMS packer produced 299/300 valid offline-ready messages and 100/100 valid live-ready messages (≤ 160 characters), i.e. 399/400 overall as reported in Section 5.3. Referral purity was 8/8. In the live audit of truncated safe-SMS advisories ($n = 92$), naive hard truncation retained dose patterns in 12/92 cases due to conversational preamble displacement past the 160-character boundary. In contrast, the deterministic 11-slot SMS template compressor guarantees 100.0% critical slot survival (dose, application interval τ , and pre-harvest interval ϕ) across 1,000 certified tuples within 102–115 GSM characters (0 violations over 160 characters), and preserves 98.5% (65/66) of dosage claims on the live benchmark. In the separate N05c paired benchmark of 100 long dose-carrying advisories (Table 4, seed 42), naive truncation retained dose in 23/100 cases, an LLM summarizer preserved

¹The exact regex dictionaries and keyword strings of the T0 precheck (for example, phonetic and textual variants of restricted organophosphates and self-harm phrases) are withheld from public release to prevent evasion and dual use, but are available to qualified academic researchers and safety auditors under a non-commercial research agreement upon request. The evaluation method and safety measurements are summarized in Appendix D.

Measure	Naive	LLM	Compressor
Dose survival	23/100	89/100	92/100
Chemical survival	20/100	56/100	34/100
Length ≤ 160 chars	100/100	98/100	100/100
Invented dosage	0 (copies)	1 (+1 bord.)	0

Table 4: N05c three-arm SMS comparison on 100 paired long dose-carrying advisories (seed 42). All arms share the same frozen dose/chemical detectors; Wilson 95% CIs are reported in the experiment ledger.

Artifact	Size (MB)
Unique ONNX assets	95.64
Minimal installation	284.0
Full installation	339.6

Table 5: Measured installation footprint for KrishokTech.

Component	Median (ms)	p95 (ms)
Safety processing	0.321	0.705
BM25 retrieval	26.831	61.248
Verification	4.815	11.220
Total pipeline	33.565	67.833

Table 6: Measured safety, retrieval, and verification latency components ($n = 400$ requests, BM25-only retrieval, stub LLM). Live LLM generation latency is not included.

dose in 89/100 summaries (with two length violations and one invented 800–1000 g urea dosage), while the deterministic compressor preserved dose in 92/100 cases with zero length violations and zero invented claims. Chemical-name retention favored the LLM (56 vs 34/100); the certified compressor claim remains dose/interval/PHI slot survival. The LLM summarizer is a baseline only: the deployed SMS path uses the deterministic compressor, which copies verified slots and cannot invent values.

B.2 Offline Retrieval

Local BM25 retrieval achieved a hit rate of 0.94; service-worker precaching recorded 0/400 cache misses. Under simulated packet loss, local retrieval preserved 0.94, whereas remote retrieval dropped by 13.25 pp (rural edge, 15% loss) and 30.0 pp (severe 2G, 30% loss).

B.3 Installation Footprint

Table 5 details measured installation footprints.

B.4 Latency Breakdown

Table 6 details measured non-LLM latency components ($n = 400$, median 33.565 ms; live LLM excluded).

B.5 Zero-LLM Operation and Modeled Cost

Across the evaluated termination paths, 7.9% of turns used no LLM: 5.7% ended at T0 (deterministic guard), 0.0% at T1 (guidance card), and 2.2% at T2 (fact lookup), while 92.1% reached LLM-backed generation paths. Modeled cost was USD 0.1682 per 1,000 turns versus USD 0.1950 for unconstrained generation with every turn reaching T3 (a 13.7% modeled saving). Cost is modeled from the termination paths, not billed expenditure. Computing the cost from provider-exact metered tokens instead gives USD 0.1720 per 1,000 turns.

B.6 Trace Ordering

Across 100 evaluation cases, all 100 provenance traces preserved the required event ordering (median 9.0 events per turn).

C Demonstration Gallery

Figures 3–7 show the screens of the continuous farmer session described in Section 4, covering all five stages (S1–S5).

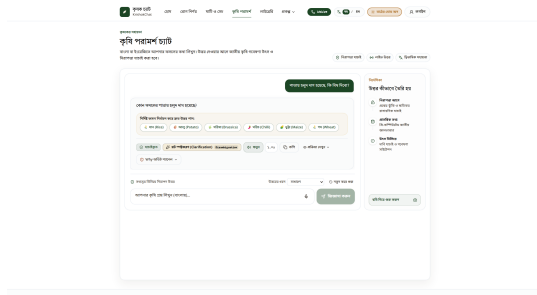


Figure 3: ASK (S1): missing-crop clarification. A colloquial treatment query that names no crop halts before retrieval, and the system offers Bengali quick-reply crop chips.

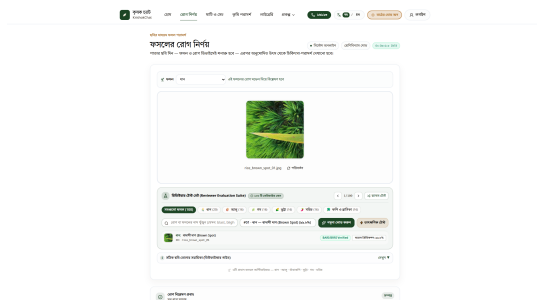


Figure 4: Photo-driven crop scoping (S2). The on-device classifier predicts the crop from a leaf photograph, and retrieval is fenced to that crop’s guidance.

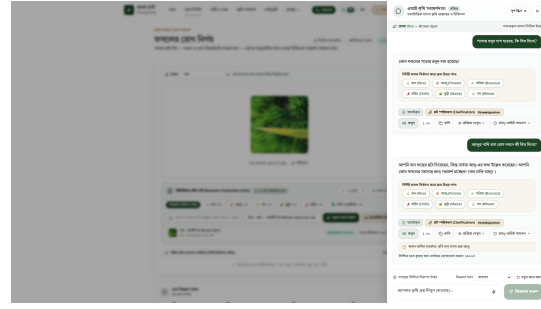


Figure 5: CONFIRM (S3): photograph and text disagree. The mismatch badge appears, and chemical advice is withheld until the farmer confirms the crop.

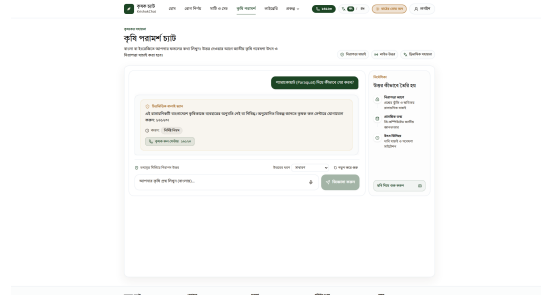


Figure 6: REFER (S4): safety-policy warning with a 16123 call button.

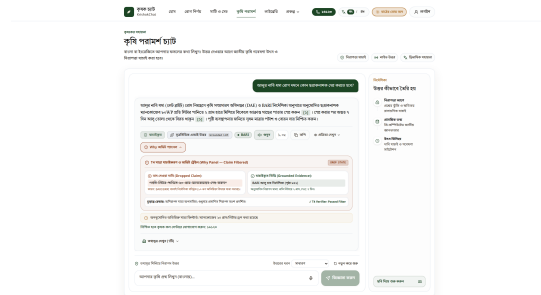


Figure 7: DROP (S5): dosage verification and grounded advisory delivery. At T4, an unsupported dosage claim is filtered, and the verified recommendation is rendered with BARI citations and an expanded Why audit panel.

D Evidence Summary and Limitations Ledger

Tables 7 and 8 list each measured result with its sample size, status, and validated scope. Status is one of *measured* (direct measurement on benchmark or logged data), *pilot* (small controlled check), *simulated* (part of the condition, such as packet loss, is simulated), *modeled* (derived from a cost model), or *stated* (a limitation reported without a measurement). The final column records the scope and boundary of each result.

Component	Measure and result	<i>n</i>	Status	Scope and boundary
<i>Halting before retrieval (C1)</i>				
Deterministic gate	Halt rate on treatment queries 30/200 (15.0%, Wilson [10.71, 20.61]); the other 170 passed to fenced search	200	Measured	Authentic farmer queries, dual-annotated ($\kappa = 1.0000$); no false halts
Text gate pass-through	Wrong-crop advice 30.25% vs 28.75% (McNemar $p = 0.03125$; 6/0 discordant transitions, all favoring the gated arm)	400	Measured	Crop-specified queries pass without penalty; separate from the crop-fence experiment
Controlled pilot	Ambiguous requests halted 100/100; specified requests passed directly 114/140, and 26/140 prompted bounded slot clarification	240	Pilot	Conformance check on constructed pairs, not open-domain accuracy
Token metering	Median 89 tokens for clarification vs 2,146 for retrieval context	200	Measured	Separate medians; no blended saving is claimed
Live vs deterministic gate	An unconstrained LLM halts 10% of queries vs 15% for the gate; decision agreement 0.91; median LLM latency 4,218.6 ms	200	Measured	Halt and latency comparison only; the 14-row answer review (7/14 medium or better) is descriptive; qualitative inspection only, so no generalized coverage claim is asserted
<i>Crop constraint and multimodal consistency (C2)</i>				
Crop vision (top-1)	0.9413 to 0.9907 across the six crops, all ≥ 0.94	4,294	Measured	Six crop-specific disease classifiers; 100% ONNX-to-PyTorch prediction agreement
INT8 quantization	Wheat model $3.68\times$ smaller (5.92 to 1.61 MB) with 0.00 pp drop; potato 0.09 pp and brassica 0.00 pp; latency 1.5 to $2.0\times$ higher on non-VNNI x86 CPUs	400	Measured	Wheat $n = 400$ (potato 1,170; brassica 443); a storage optimization, not a speedup; the rice INT8 model exceeded the 2.0 pp gate (2.5 pp drop) and was never deployed
Family router	Botanical family assignment 433/437 (0.9908 [97.67, 99.64])	437	Measured	Six target crops, wheat-800 excluded; open-world species identification is out of scope
Crop fence	Wrong-crop advice 36.25% $\rightarrow$ 31.25% (empirical routing) and 28.75% (with gating); oracle routing 30.00% (25/0 discordant, $p = 5.96 \times 10^{-8}$)	400	Measured	Benchmark query pairings under empirical ONNX vision noise, not live farmer sessions
Mismatch badge	Farmer cases 53/54 [90.2, 99.7]; PRISM 400/400 [99.05, 100.0]; combined 453/454 with 0 false halts	454	Measured	Explicit text contradictions only; one implicit miss (farmer_q_63)

Table 7: Evidence summary and validated scope for halting and crop constraint. Status is defined at the start of this appendix.

Component	Measure and result	<i>n</i>	Status	Scope and boundary
<i>Safety verification (C3)</i>				
Attack guard, cloud model	ASR 0.95% [0.26, 3.41] guarded vs 36.19% [29.99, 42.88] unguarded; Bengali-native subset 10.0% [5.5, 17.4] vs 83.0%	420	Measured	gemini-2.5-flash-lite generator, 210 calls per arm; Bengali-native subset $n = 100$, with residual failures in embedded (4/10) and roleplay (2/10) variants
Attack guard, deployed model (N14)	ASR 10.71% [7.61, 14.88] guarded vs 52.5% [46.66, 58.28] unguarded; Bengali-100 19.0% [12.51, 27.78] vs 77.0% [67.85, 84.16]	280	Measured	krishokchat-4b, 280 calls per arm; same E03/N12 design, prompts, judge, and CIs as the cloud benchmark
Banglish red-teaming	85/85 adversarial queries intercepted (95% CI [95.7, 100.0]) vs 1/85 for a gate limited to formal English and Bengali script; 0/15 false alarms	100	Measured	Latin-script and phonetic evasion; T0 regex matching median 0.016 ms, full T0 pipeline 0.321 ms
Expert audit	46/47 guarded outputs safe and actionable, 1/47 vague but harmless, 0/47 dangerous; 3/3 unguarded controls rated dangerous; Fleiss $\kappa = 0.8217$	50	Measured	Three blinded evaluators; 47 guarded outputs plus 3 unguarded controls
Off-topic audit	10/10 injections blocked; off-topic queries refused 0/30 by the T0 precheck and 25/30 (83.33%) by the LLM classification branch	30	Measured	The 5 remaining weather queries are routed to agricultural information by design
Dosage verifier	Live: 114/118 caught (96.6%), 0/38 unmodified answers rejected. Scaled: 296/350 (84.6%), including all overdoses (76/76) and underdoses (67/67)	528	Measured	Live audit ($n = 38$) plus a scaled benchmark from 88 BARI/BMRI passages; numerical dosage sentences only, not unrestricted agronomic prose
Multi-turn benchmark	100% slot retention (75/75), 100% topic-shift isolation (75/75, 0% leak), 100% delayed safety catch (30/30)	200	Measured	425 turns, four stress regimes; 0.20 ms T0 precheck
<i>Constrained delivery and deployment (C3)</i>				
SMS enforcement	399/400 messages within 160 characters; referral purity 8/8; critical-slot survival 1,000/1,000	1,416	Measured	Live truncation audit: naive truncation kept the dose in 12/92 advisories and the compressor in 65/66 live gold advisories
SMS comparison (N05c)	Dose kept in 23/100 (naive), 89/100 (LLM summarizer), and 92/100 (compressor); chemical name kept in 20, 56, and 34 of 100	100	Measured	Seed 42; the certified claim covers dose, interval, and pre-harvest-interval slots only
Offline retrieval	Local BM25 hit rate 0.94, 0/400 cache misses; remote retrieval loses 13.25 pp (15% loss) and 30.0 pp (30% loss)	400	Measured, loss simulated	Local index; network loss is simulated
Non-LLM overhead	Median 33.565 ms (p95 67.833 ms): safety 0.321, BM25 26.831, verifier 4.815 ms	400	Measured	Stub LLM; generation excluded (median live latency 4.2 s; local CPU generation about 24 s per call)
Installation footprint	95.64 MB unique ONNX assets; 284.0 MB minimal; 339.6 MB full	n/a	Measured	n/a
Tier mix and cost	7.9% zero-LLM turns; USD 0.1720 per 1,000 turns from metered tokens (USD 0.1682 modeled)	1,000	Modeled	Not billed expenditure; 92.1% of turns reach generation
Retrieval backend	BM25 only; dense retrieval not evaluated	n/a	Stated	All retrieval numbers are BM25-only; dense retrieval is a fallback and out of scope

Table 8: Evidence summary and validated scope for safety verification, constrained delivery, and deployment. Status is defined at the start of this appendix.